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E. Split

time limit per test: 2 seconds
 memory limit per test: 256 megabytes

Farmer John has an array a containing n positive integers and an integer k .

Let $a[l, r]$ be a subarray* of a . He performs the following procedure to independently determine if subarray $a[l, r]$ is awesome:

- Initially, FJ has k empty **multisets**, numbered from 1 to k .
- Then, for each element a_i ($1 \leq i \leq n$) in a :
 - If $l \leq i \leq r$ (that is, a_i is in the subarray $a[l, r]$), he places a_i in multiset 1,
 - Otherwise, he places a_i into any multiset he wants (**which may be multiset 1**).
- Subarray $a[l, r]$ is awesome if there is some way for him to place elements such that, for every value v , all multisets contain the same number of elements with value v . In other words, he wants to make all multisets contain the exact same elements (ignoring ordering).

Output the number of awesome subarrays.

*For array a of size n and integers $1 \leq l \leq r \leq n$, the subarray $a[l, r]$ denotes the array consisting of the elements $a_l, \dots, a_r$, in order.

Input

The first line contains an integer t ($1 \leq t \leq 1000$) — the number of test cases.

The first line of each test case contains two integers n and k ($2 \leq k \leq n \leq 2 \cdot 10^5$).

The following line contains n space-separated integers $a_1, a_2, \dots, a_n$ ($1 \leq a_i \leq n$).

It is guaranteed that the sum of n over all test cases does not exceed $2 \cdot 10^5$.

Output

For each testcase, output one integer on a new line: the number of awesome subarrays.

Example

input	Copy
4 3 2 1 1 1 4 2 1 2 1 2 8 2 3 3 3 3 2 2 2 2 6 3 1 1 1 1 1 1	
output	Copy
0 7 18 11	

Note

Test case 1: $n = 3$, $a = [1, 1, 1]$.

For $k = 2$, you cannot finish with the same number of 1's in both multisets, so no subarray works.

Test case 2: $n = 4$, $a = [1, 2, 1, 2]$.

Codeforces Round 1050 (Div. 4)

Contest is running

01:14:59

Contestant



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Language: Python 3.13.2
 Almost always, if you send a solution on PyPy, it works much faster

Choose file: Choose File No file chosen

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For $k = 2$, the final state must give each multiset exactly one 1 and one 2. Thus a valid subarray can contain at most one 1 and at most one 2.

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